



Authorship Arabic texts using Deep Impostors method

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PROBLEM DEFINITION

Medieval Arabic texts present significant authorship verification challenges due to complex language, orthographic variation, and limited labeled data. This project proposes an automated approach to distinguish authentic and disputed texts.

PROJECT GOAL

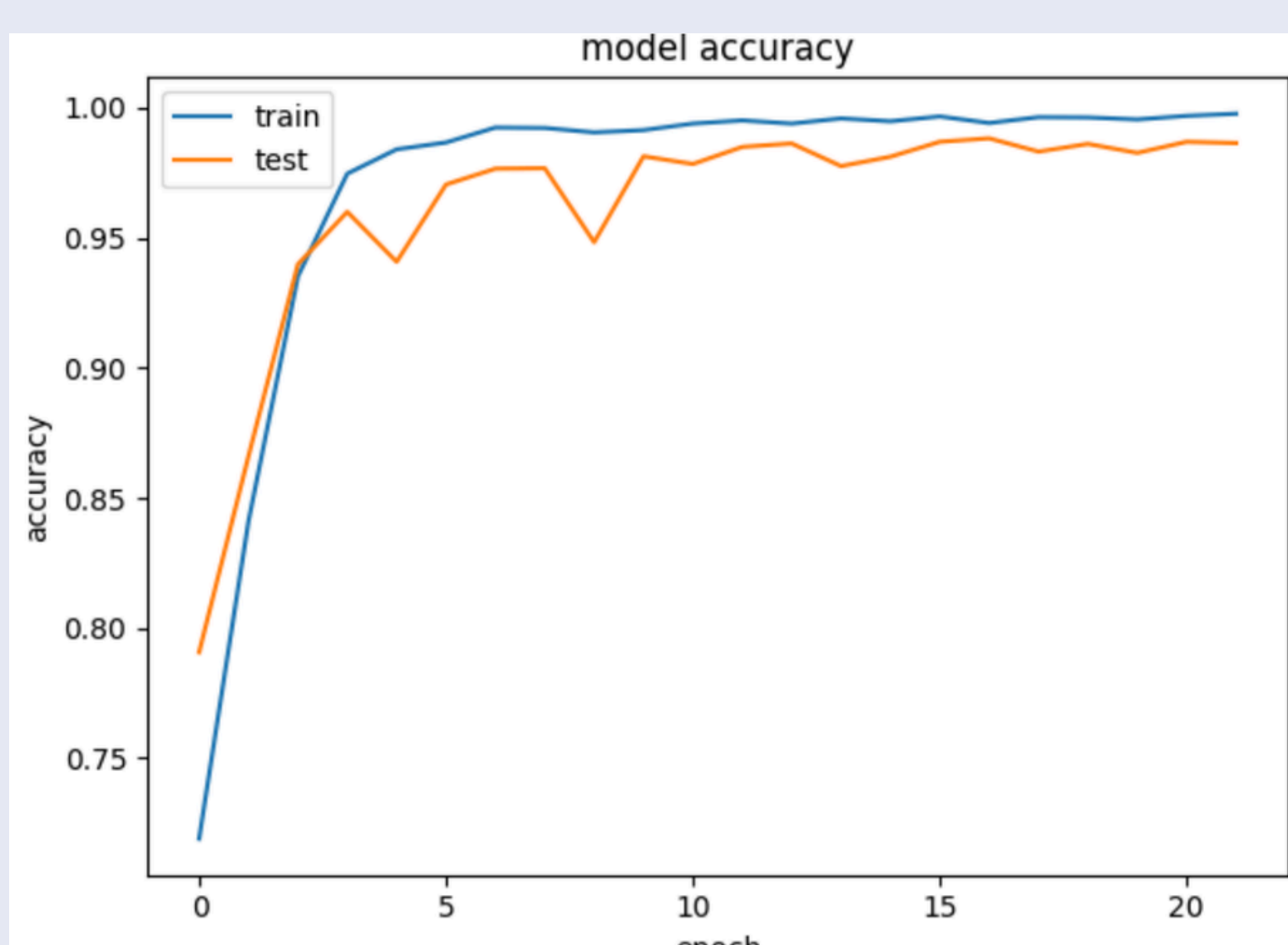
The goal is to develop an unsupervised system for Arabic authorship verification that identifies stylistic consistency and detects disputed texts using deep learning and similarity analysis.

SOLUTION

An automated framework for Arabic authorship verification of Al-Mutanabbi's poetry was developed. The proposed system identifies authentic and potentially disputed poems based on their writing style, providing an effective tool for computational literary analysis.

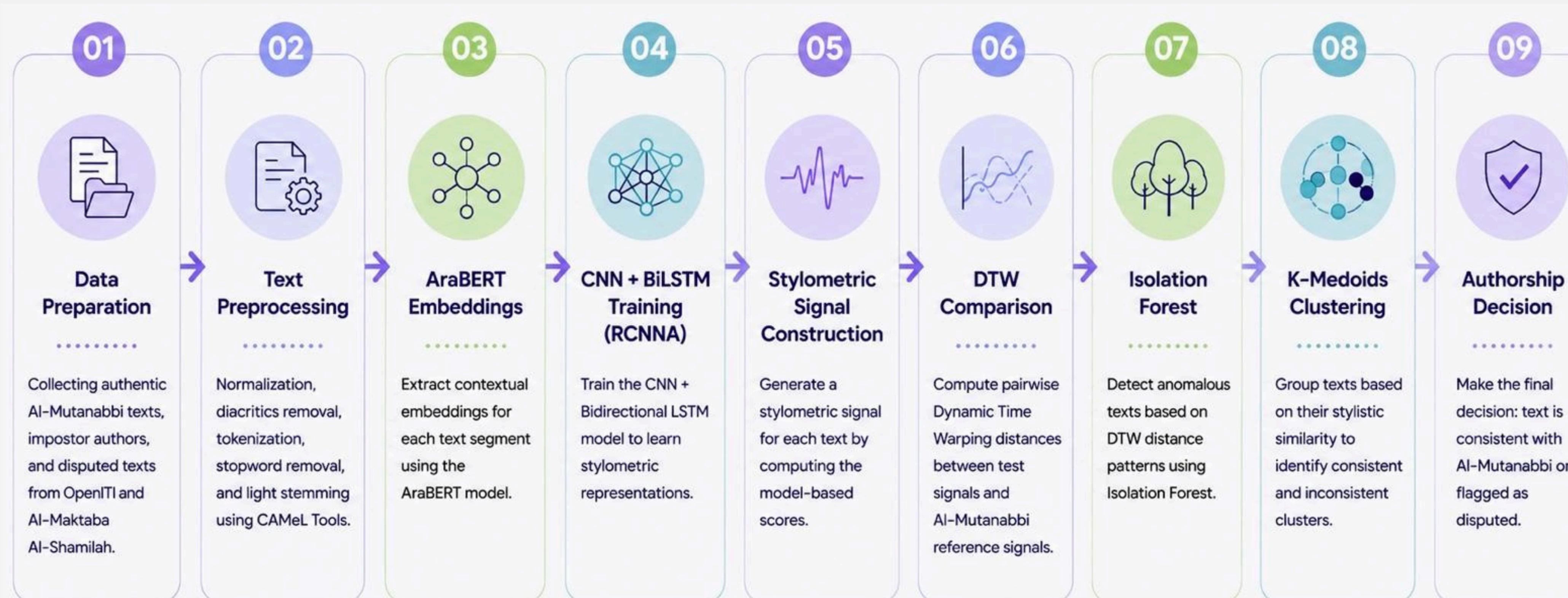
DATA COLLECTION

We collected 300 poems by Al-Mutanabbi from Al-Diwan and compiled texts from 25 impostor authors to build the authorship verification dataset.

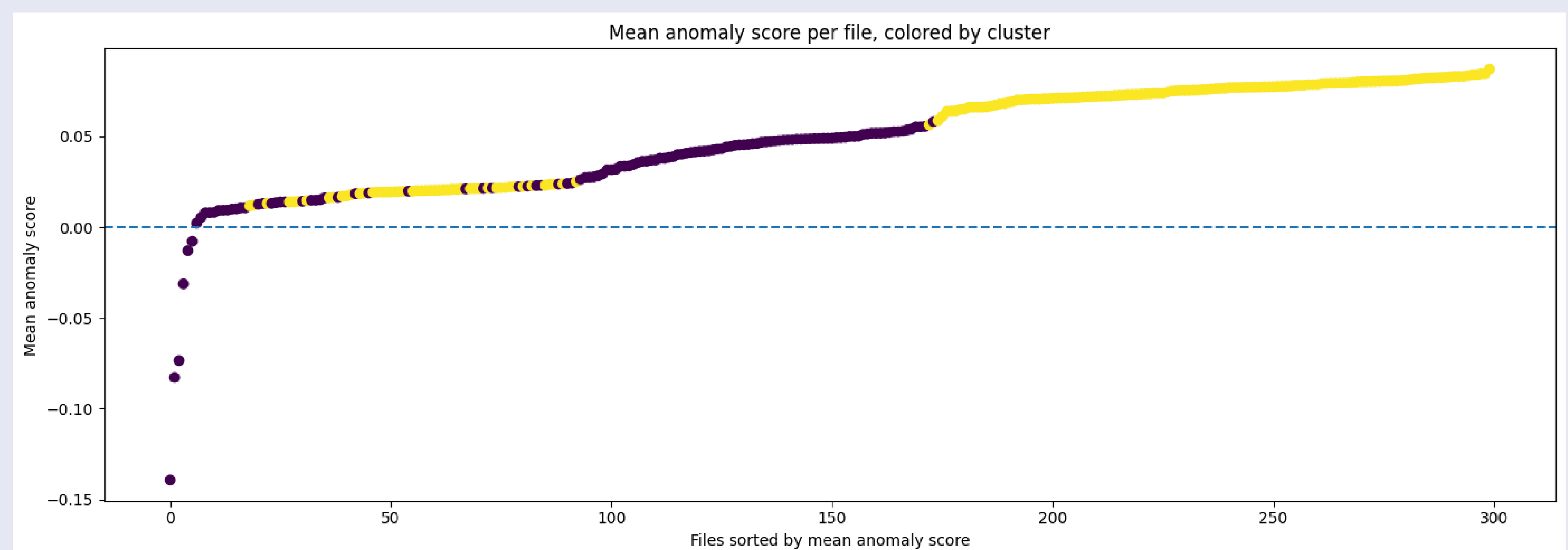


Training and test accuracy improved consistently over the training epochs, reaching approximately 99%. The close agreement between both curves indicates stable learning and good generalization.

PIPELINE



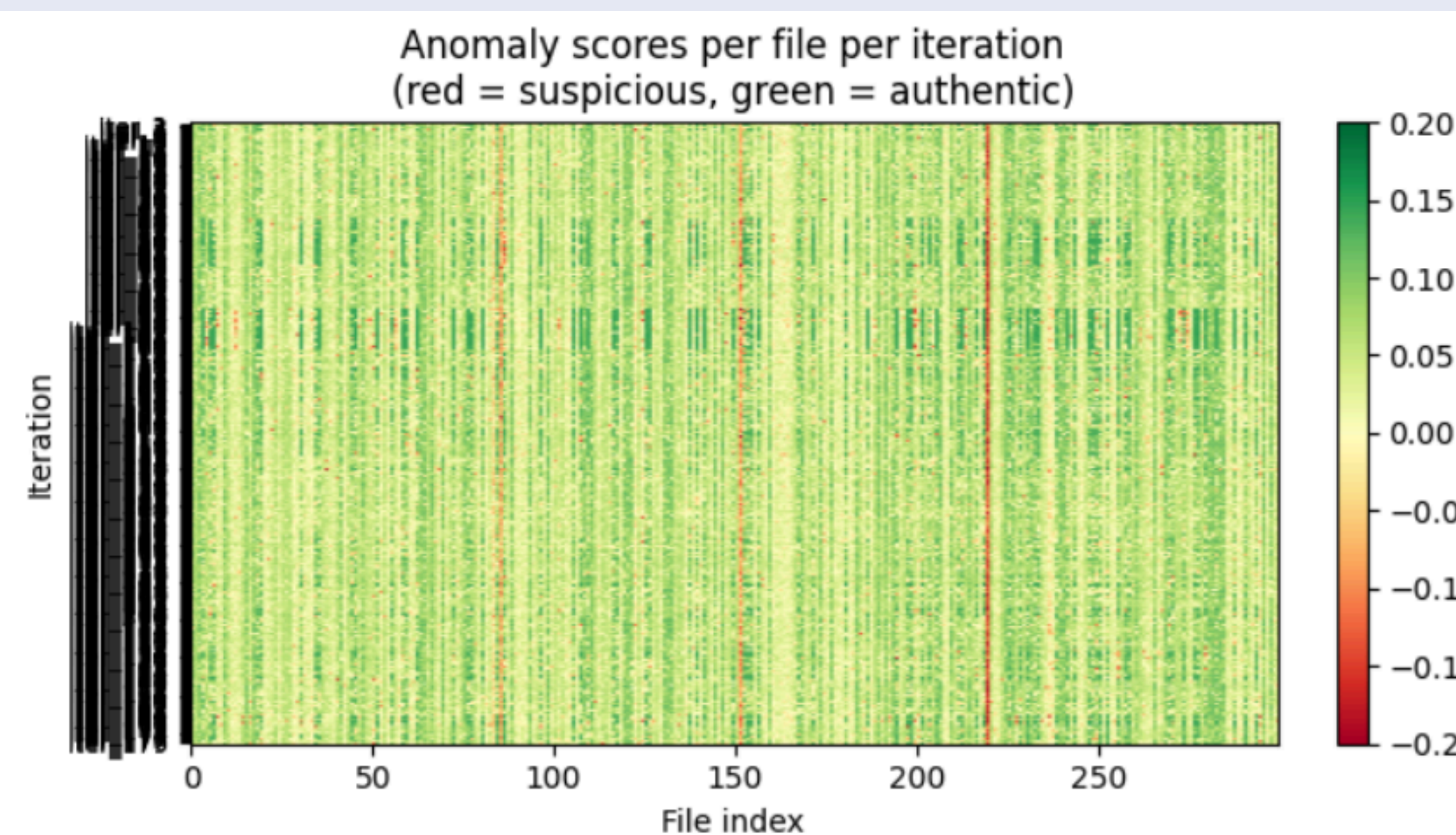
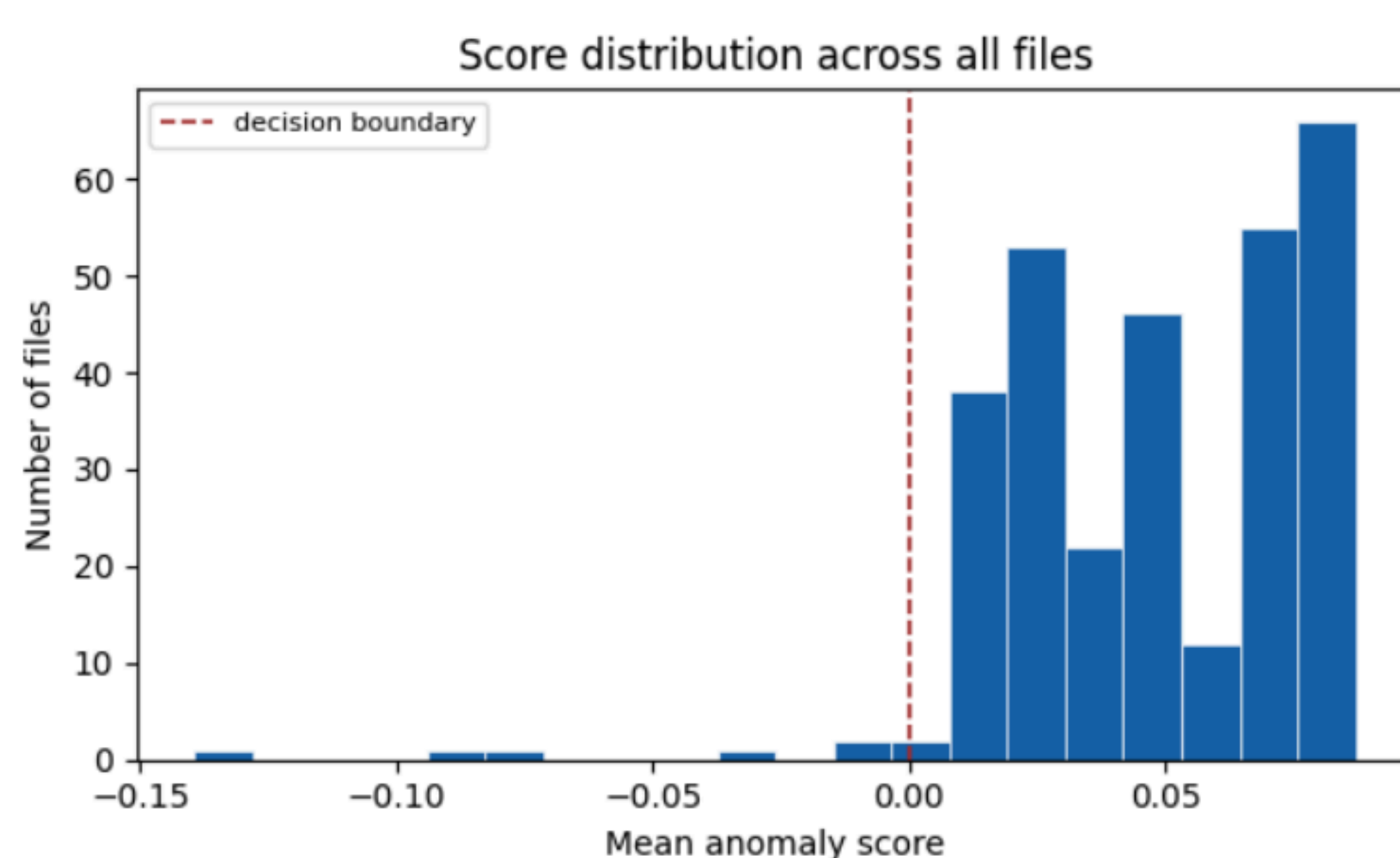
RESULTS



This figure presents the mean anomaly score of each analyzed text after authorship verification. The predominance of positive scores indicates that most texts are stylistically consistent with Al-Mutanabbi, while only a few outliers are identified as anomalous.

CONCLUSION & FUTURE WORK

Most texts attributed to Al-Mutanabbi were identified as authentic, while only a small number were detected as potentially disputed. **Future work** includes expanding the dataset and evaluating additional transformer models to further improve authorship verification performance.



The heatmap presents the anomaly score of each file across all iterations. Most files remain consistently authentic (green), while a small number of files exhibit suspicious scores (red) in one or more iterations.

